Yeonsu Jung

Physics-based simulation · X-ray CT and 3D imaging · Experimental mechanics · Contact-rich robotics

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SUMMARY

Mechanical engineer (Ph.D.) and Research Associate at Harvard School of Engineering and Applied Sciences. I work on contact-rich and deformable systems in three ways: physics-based simulation and the numerical optimization that runs on it; X-ray micro-CT and 3D image analysis, where I build the fixtures, run the scanners and write the analysis; and experiments. What I most want to work on is real-to-sim-to-real transfer: a simulator built from measurements of the real system, calibrated against hardware, and results that hold when they go back to it. I run multiple projects of more than 100 commits each with AI coding and science agents (Claude Code, Codex), every step gated by reproduction and conformance checks. Co-author of 14 peer-reviewed journal papers, five of them in *PNAS*, and others in *Science Robotics* and *Science Advances*.

SKILLS

Simulation	MuJoCo (C++ plugin API, MJX, MuJoCo Warp, mjlab), NVIDIA Warp; my own rigid-body contact engine in C++17 with OpenMP and CUDA; discrete elastic rods; complementarity and penalty contact with friction; finite elements (COMSOL)
Optimization	Differentiable simulation in JAX (`jit`, `vmap`, `scan`); trajectory optimization; sparse Newton and Gauss–Newton; FIRE on GPU; augmented Lagrangian; CasADi/IPOPT; control and learning: PPO (RSL-RL), LQR, receding-horizon control
X-ray CT, imaging	Micro-CT on a Nikon X-Tek HMXST 225 and a Zeiss Xradia Versa 620; fixtures and in-situ load frames; 3D segmentation and object tracing; reconstruction (ASTRA); Dragonfly (Python), VGStudio, ParaView, Blender; OpenCV tracking; high-speed imaging; PIV
Fabrication, testing	Fixture and apparatus design; certified user of the Harvard SEAS machine shop (lathe, mill, bandsaw, grinder); CAD in SolidWorks, Fusion and Onshape; 3D printing (Prusa, Formlabs); molds and polymer casting; bending, centrifugal and load-frame tests
Software	Python, C++17, CUDA, JavaScript, MATLAB; PyTorch; pybind11, CMake, pytest, GitHub Actions, SLURM; Claude Code, Codex

SELECTED PROJECTS

Entangled rod packings: optimization, contact engines, X-ray CT — 2021–2026 · PNAS 2025 · PRL under revision

- Wrote a JAX optimizer (FIRE, on GPU) that maximizes the entanglement of up to 2,000 rods without overlap, and a C++17 engine for rigid rods in contact (cone complementarity by projected Gauss–Seidel, soft contact, OpenMP, a CUDA broad phase).
- Showed that maximally entangled packings cage themselves and hold together by friction alone. Re-ran the dynamics in MuJoCo to test which results survive a change of contact model: the cooling law does (time constant within 15%), contact lifetimes do not.
- Measured real steel-rod packings by X-ray micro-CT: built the fixtures, ran the scans, and wrote a pipeline that traces about 4,600 rods in a 2000 × 2000 × 741 voxel scan in 12 minutes.

X-ray micro-CT and 3D image analysis — 2021–2026 · eLife 2025

- Scan metal, polymer, gel, granular and biological samples on two systems, and write the software that turns the scans into measurements. A browser viewer ray-marches five of the tomograms on the GPU.
- Building in-situ load frames for both scanners (ongoing): force is logged while a packing is scanned at each load step, and each scan gives its contacts and the contact network's algebraic connectivity.
- When scans on a shared scanner began to show out-of-focus halos, ruled out the sample with a series of test scans and showed that the X-ray beam and the detector were misaligned; the service company then fixed it on site.

Elastic rods in joint coordinates, for robot simulators — 2025–2026

- Derived one swing–twist rod energy with an exact ball-joint torque, implemented four ways (NumPy, JAX, PyTorch in mjlab, a C++ MuJoCo plugin) that agree to 2×10^{-11}. Found that MuJoCo's stock cable torque is not the gradient of an energy (off by 7.7–46.7% once the bend axis tilts).
- Control on the verified model, in simulation: a whip crack optimized by differentiating through MJX (tip at 68 m/s, 5.8 mm from its target); a periodic LQR; PPO in 4,096 parallel environments (MuJoCo Warp) that rears a 16-link chain; receding-horizon control that rights a tumbling glider in 10 of 10 launches.

Models of collaborators' experiments — 2022–2026 · PNAS 2022 · PNAS 2026

- Calibrated a simulator of centimeter-scale vibration-driven robot chains against hardware: it matches 23 of 30 cells of the measured stability map. I rejected a variant that fitted better but stretched the joints beyond their measured play.
- Predicted how rotationally 3D-printed lattices morph when heated, with discrete elastic rods, and fitted designs to target surfaces in JAX (Lewis group); analysis and micro-CT of an entanglement gripper (Wood group; reproduced N. Charles's simulation).

EXPERIMENTS

- **Swallow nests** (*PNAS* 2021, first author): measured the strength of real nests and of polymer-bridged granules in three-point bending and in a centrifuge, printed nests from clay and mucin, and computed the stress in a 3D-scanned nest by finite elements (COMSOL): it peaks where a real nest tore away.
- **Hydrogel beads on a hot pan** (*Physical Review E*, under revision, corresponding author): devised the Leidenfrost experiment with M. Cho and J. Ha, in which a bead gets its energy only from vapor bursts at the wall, and model its motion.
- **Earlier:** artificial roots drawing water from packed beads (*J. R. Soc. Interface* 2019); irregular cylinders rolling down a ramp, tracked on video (*PNAS* 2025).

EXPERIENCE

Research Associate in Applied Mathematics 2026–present
Postdoctoral Fellow 2021–2026
Harvard John A. Paulson School of Engineering and Applied Sciences, Cambridge, MA · with L. Mahadevan

- Simulation, X-ray CT and experiments on contact-rich filamentary systems: collision handling, nonlinear rod mechanics, 3D image analysis, inverse design. Work with experimental groups in robotics, materials and 3D printing; mentor junior researchers.

Postdoctoral Fellow 2020–2021
Rowland Institute at Harvard, Cambridge, MA

- Modeling, fabrication and flow measurement (particle image velocimetry) of shark-skin-inspired drag-reducing surfaces.

Postdoctoral Fellow 2019–2020
Seoul National University, Seoul, South Korea

- Finite-element (COMSOL) computation of the electric field around ionic spiderwebs (*Science Robotics* 2020) and the model of soft electroreceptors (*Science Advances* 2021); mechanical tests and finite-element analysis of mud nests (*PNAS* 2021).

EDUCATION

2019 **Ph.D., Mechanical Engineering**, Seoul National University (advisor H.-Y. Kim). Thesis: Hydrodynamic principles of water uptake of plant roots.
2014 **M.S., Biomedical Engineering**, Gwangju Institute of Science and Technology (GIST)
2012 **B.S., Mechanical Engineering**, Pohang University of Science and Technology (POSTECH)

SELECTED PUBLICATIONS

Y. Jung, L. Mahadevan. Emergent cohesion via self-caging in maximally entangled rod packings. *Physical Review Letters*, under revision. arXiv:2606.03952 (2026).

M. Cho, **Y. Jung**, J. Ha. Boundary-driven activity of a Leidenfrost hydrogel. *Physical Review E*, under revision (2026).

M. K. Abdelrahman, J. K. Wilt, **Y. Jung**, et al. Rotational 3D printing of active–passive filaments and lattices with programmable shape morphing. *PNAS* 123, e2537250123 (2026).

Y. Jung, T. Plumb-Reyes, H.-Y. G. Lin, L. Mahadevan. Entanglement transition in random rod packings. *PNAS* 122, e2401868122 (2025).

S. Yin, C. Liu, G. P. T. Choi, **Y. Jung**, et al. Morphogenesis and morphometry of brain folding patterns across species. *eLife* 14, RP107138 (2025).

K. Becker, C. Teeple, N. Charles, **Y. Jung**, et al. Active entanglement enables stochastic, topological grasping. *PNAS* 119, e2209819119 (2022).

W. J. Song*, Y. Lee*, **Y. Jung***, et al. Soft artificial electroreceptors for noncontact spatial perception. *Science Advances* 7, eabg9203 (2021). *Equal contribution.

Y. Jung, S. Jung, S.-i. Lee, W. Kim, H.-Y. Kim. Avian mud nest architecture by self-secreted saliva. *PNAS* 118, e2018509118 (2021).

Full list of 14 journal papers: yeonsu-jung.github.io/publications.html

SELECTED TALKS

2026 Condensed Matter Theory Kids' Seminar, Harvard
2025 Physics of Living Systems, MIT
2024 Network Science Institute, Northeastern University
2023 Physics colloquia, Michigan Technological University and Bard College; APS March Meeting

AWARDS AND TEACHING

2021–22 Postdoctoral Fellowship Abroad, National Research Foundation of Korea
2014–16 Superior Academic Performance Scholarship, Seoul National University
2008–10 National Scholarship, National Research Foundation of Korea
2024 Teaching Fellow, Gen Ed 1190, Harvard University